

QUALITY ASSURANCE

Zerado landing page — QA report (accessibility & cross-browser lane)

ForgePlay deliverable · engineered artifact

FlowForge ENGINEERED ARTIFACT		DRAWING TITLE Zerado landing page — QA report (accessibility & cross-browser lane)		
DISCIPLINE	SHEET	DWG NO.	REV	STATUS
QUALITY ASSURANCE	01 OF 13	FP-LP-QA-01	A · 2026-08-25	● IN REVIEW
LEGEND Field — blueprint Ink — annotation Accent — registration Pass — verdict				
Rendered from qa-report.md			powered by FlowForge	

Verdict: FAIL — 1 blocking finding.

Artifact	<code>http://localhost:4321/</code> (Astro preview of <code>site/dist/index.html</code> , 52,880 bytes)
Lane	accessibility + cross-browser. Core Web Vitals / bundle size deliberately NOT run — that is the <code>performance</code> lane's scope.
Tooling	Playwright 1.57 (<code>qa/harness/</code>), Chromium 151.0.7922.34 · Firefox 153.0 · WebKit 26.5; <code>@axe-core/playwright</code> + axe-core 4.11
Raw data	<code>qa/qa-raw-results.json</code> (axe + overflow, 4 viewports) · <code>qa/qa-crossbrowser.json</code> (3 engines × 4 viewports)
Screenshots	<code>qa/screenshots/</code> — 30 files

Findings: 1 BLOCKING · 2 MAJOR · 7 MINOR.

The page is, with one exception, unusually clean: **zero axe violations at all four viewports**, zero contrast failures across 206 rendered text elements, a correct skip link, a complete 16-section DOM in ratified order, a fully honest content sweep, and near-identical rendering across three engines. The single blocking defect is a flex-sizing bug in §02 that **cuts a ratified sentence in half on mobile** — and it is invisible to the exact gate assertion the blueprint prescribed, which is why it survived to QA.

FlowForge	DRAWING TITLE					● IN REVIEW
	Zerado					
	landing page					
	— QA report					
	(accessibility					
	& cross-					
	browser lane)					
SHEET		DWG NO.		REV		
02 OF 13		FP-LP-QA-01		A · 2026-08-25		

§01 Findings

🔴 BLOCKING-1 — §02 `maker-line` text is clipped at 375 and 768; the FlowForgeSoft attribution is unreadable on mobile

What. The paragraph `.z-maker-line__text` has an intrinsic width of **762px** and **never shrinks or wraps**. Its flex row is `flex-wrap: nowrap` and the paragraph is `flex: 0 0 auto`, so at narrow viewports it simply runs off the side of the page and is clipped by `overflow-x: hidden` on `<html>` / `<body>`. The text is not scrollable-to; it is **destroyed**.

Where. `#maker-line > .z-maker-line__row > p.z-maker-line__text` — at 375px and 768px. Reproduces **identically in Chromium, Firefox and WebKit**.

Evidence (`qa/harness/mk2.mjs`, character-level `Range.getBoundingClientRect()` measurement of what actually falls inside the viewport):

```
= 320px = text right=806 OVERSHOOT past viewport = 486px  clipped
characters: 50
FULL    : "Zerado is the second product from FlowForgeSoft, the
company behind FlowForge."
VISIBLE : "Zerado is the second product"
= 375px = text right=806 OVERSHOOT past viewport = 431px  clipped
characters: 45
VISIBLE : "Zerado is the second product from"
= 768px = text right=814 OVERSHOOT past viewport = 46px   clipped
characters: 5
VISIBLE : "Zerado is the second product from FlowForgeSoft, the
company behind FlowF"
= 1280px = OVERSHOOT = -210px                                clipped
characters: 0
```

Cross-engine (`qa/harness/xb-maker.mjs`):

```
chromium @375: OVERSHOOT=431px @768: OVERSHOOT=46px
firefox  @375: OVERSHOOT=431px @768: OVERSHOOT=46px
webkit   @375: OVERSHOOT=431px @768: OVERSHOOT=46px
```

Screenshots: `screenshots/DEFECT-maker-line-375.png` (reads *"Zerado is the second product from F"*), `DEFECT-maker-line-768.png` (reads *"...the company behind FlowF"*), plus `-320`, `-1280`, and `-firefox-375/768`, `-webkit-375/768`.

Why it matters.

1. **It deletes ratified content.** `decisions.md` records permission to name FlowForgeSoft as an answered-and-closed item, and the blueprint gives §02 an entire section to carry it. At 375 the visitor never sees the words "FlowForgeSoft" or "FlowForge" — the section renders a truncated fragment that says nothing. The one job of §02 fails on the most common viewport class.
2. **WCAG 2.1 SC 1.4.10 Reflow (AA) failure.** At 320px CSS width, content is lost with no horizontal scroll available to recover it. Clipping is content loss, not reflow.
3. It is not a cosmetic near-miss — 45 of 78 characters (58%) are gone at 375px.

Root cause is one declaration: the paragraph is `flex: 0 0 auto` inside a `nowrap` row. Changing it to permit shrinking (`flex: 0 1 auto` / `min-width: 0`) or allowing the row to wrap fixes both this and MAJOR-2. (*Reported, not fixed — I do not modify site/.*)

● MAJOR-1 — The blueprint's prescribed overflow assertion cannot detect BLOCKING-1

What. Blueprint §5.5 specifies the gate assertion:

`document.documentElement.scrollHeight` `document.documentElement.clientWidth`. That assertion **passes at all four viewports** — while content is being clipped.

Evidence (`qa/qa-raw-results.json`):

```
= 375 = docEl 375/375 pass=True | body 806/375 | elements past  
viewport: 3  
= 768 = docEl 768/768 pass=True | body 814/768 | elements past  
viewport: 3  
= 1280 = docEl 1280/1280 pass=True | body 1280/1280 | elements past  
viewport: 1  
= 1920 = docEl 1920/1920 pass=True | body 1920/1920 | elements past  
viewport: 0
```

`html` / `body` carry `overflow-x: hidden` (measured, all three engines). That pins `documentElement.scrollHeight` to `clientWidth` **by construction**, so the assertion is a tautology and can never fail.

Why it matters. §5.5 ranks its mechanisms — reflow first, reformat second, contained scroll as a "backstop only" with `tabindex="0"` so keyboard users can reach it. A blanket `overflow-x: hidden` on the root is **none of the three**: it is a fourth mechanism that silently discards content and simultaneously disables the

only automated check the design specified. Any future width regression is equally undetectable. The honest assertion is per-element (`element.right ≤ clientWidth` for non-`aria-hidden` content), which is what `qa/harness/run-core.mjs` runs and which caught this.

The only other element extending past the viewport is `.z-frame__watermark` — verified `aria-hidden="true"` decorative repeating "PHASE 4" text inside a deliberate `.z-frame__watermark-clip`. **Not a defect.**

● MAJOR-2 — §02's flanking rules render at 0px width at every viewport

What. Blueprint §1.1 assigns §02 the layout family *"Single centred rule-flanked line"*, and §6.3's MakerLine component consumes `--z-border` for those rules. Both `span.z-maker-line__rule` elements (`flex: 1 1 0%`) are starved to zero width because the paragraph consumes the entire row.

Evidence (`qa/harness/xb-maker.mjs`, all engines): `rule widths=[0,0]` at 375 and 768. At 1280/1920 (`qa/harness/mk.mjs`) both rules measure `left=284 right=284 width=0` and `left=1094 right=1094 width=0`.

Why it matters. The ratified layout family is never realised anywhere on the page — §02 renders as a bare centred line, not the specified rule-flanked one. Same root cause as BLOCKING-1; one fix resolves both. Worth its own line because fixing only the clipping (e.g. by adding `white-space: normal` alone) would leave the rules still absent.

● MINOR (7)

#	FINDING	EVIDENCE
MINOR-1	FAQ expand/collapse motion not implemented. Blueprint §7.6 motions #7/#8 specify 220ms expand / 140ms collapse. Measured <code>transition-duration: 0s</code> on both <code>.z-faq-item__summary</code> and <code>.z-faq-item__answer</code> ; <code>animation-name: none</code> . The accordion snaps open. Defensible (native <code><details></code> cannot animate without <code>::details-content + interpolate-size</code> , which measured <code>true</code> in Chromium but <code>false</code> in Firefox and WebKit) — but §7.6 claims the nine-entry table "is the complete list", and two of the nine do not exist.	<code>qa/harness/motion2.mjs</code> , <code>run-xb.mjs</code>

#	FINDING	EVIDENCE
MINOR-2	Parked scanner track count is 5, blueprint says 4. §7.6 motion #2 enumerates "TF-1/2/3 chrome bars + §15 top edge (4 instances)". Measured 6 tracks total: 1 live (<code>#hero</code>) + parked in <code>#hero</code> figure, <code>#moods</code> , <code>#terminal-first</code> , <code>#community</code> , <code>#closing-cta</code> . The extra is the §11 community panel's chrome bar. Static in both motion modes, so zero motion harm — an inventory-count drift.	<code>qa/harness/motion2.mjs</code>
MINOR-3	Footer contact link is 19px tall. <code>a[href="mailto:alex@flowforgesoft.com"]</code> measures 180×19px at both 375 and 1280 (<code>display: inline</code> , no line-box padding). Blueprint §5.4 commits to "Every interactive element is $\geq 48 \times 48$ px at every breakpoint" and "Footer links: 44px line-box". This is NOT a WCAG failure — SC 2.5.8 (AA) is satisfied via the spacing exception (nearest other target centre is 65px away, ≥ 24 px). It is a deviation from the blueprint's own stated floor, and the only target on the page below it; every other target measured ≥ 44 px.	<code>qa/harness/targets.mjs</code> , <code>nav375.mjs</code>
MINOR-4	CoverTile art variety is 4, blueprint says 12. §6.3 specifies "twelve fixed variants (§7.5)". Measured 12 tiles (<code>.z-tile</code>) drawing on 4 art variants (<code>z-tile--A/B/C/D</code>), repeated.	<code>dist/index.html</code> grep: <code>z-tile</code> count: 12, variants: <code>['A','B','C','D']</code>
MINOR-5	StateLegend resolves to no DOM node. Listed in blueprint §6.3's component table, but §05's own layout spec never places it (it specifies "StateChip × 4 in one row", which the build renders correctly with glyph + LABEL + line). Contract self-inconsistency rather than a build gap — flagging so it is closed deliberately.	<code>qa/harness/components.mjs</code>
MINOR-6	FlowForge footer mark renders at 18px, blueprint §7.8 says 28px. <code>img[src="/flowforge-logo.svg"]</code> displayed 18×18 (natural 150×150, loads fine — <code>loading="lazy"</code> , HTTP 200).	<code>qa/harness/links.mjs</code>

#	FINDING	EVIDENCE
MINOR-7	Link inventory exceeds blueprint §2.1 — which states "exactly three destinations and nine link instances ... fifteen focusable elements". Build has 10 links / 4 destinations / 16 focusables at ≥768. The extra is the <code>https://www.flowforgesoft.com</code> "Powered by FlowForge" link, which §6.3 <i>does</i> require and which my brief requires. The build is correct; §2.1's table is stale. Separately, that link uses <code>target="_blank"</code> with no external-link cue, where §2.1 reasons that GitHub should open in the same tab.	<code>qa/harness/run-content.mjs</code>

FlowForge

DRAWING TITLE

Findings

SHEET

03 OF 13

DWG NO.

FP-LP-QA-01

REV

A · 2026-08-25

STATUS

● IN REVIEW

§02 Per-checklist results

1 · Screenshots at 375 / 768 / 1280 / 1920 — done

Full-page PNGs at `qa/screenshots/{375,768,1280,1920}-fullpage.png` (chromium), plus `{firefox,webkit}-*` for all four widths — 12 full-page captures. Document heights: 375→14,926px · 768→11,700px · 1280/1920→11,442px. Also captured: `skipLink-focused-{375,1280}.png`, `reduced-motion-{reduce,no-preference}-1280.png`, `scanner-{reduce,no-preference}.png`, and 8 `DETECT-maker-line-*` crops.

2 · axe-core WCAG 2.1/2.2 AA — 0 violations at all four viewports

Tags: `wcag2a`, `wcag2aa`, `wcag21a`, `wcag21aa`, `wcag22aa`.

```
[375] violations=0 passes=28 incomplete=1
[768] violations=0 passes=28 incomplete=1
[1280] violations=0 passes=28 incomplete=1
[1920] violations=0 passes=28 incomplete=1
```

The one `incomplete` is `color-contrast` (42 nodes at 1280), meaning **axe could not compute it** — not that it passed. Breakdown of axe's own reasons:

```
x31 "Element content contains only non-text characters" (the ○ ●
● ○ ► ■ glyphs, all aria-hidden="true")
x10 "background color could not be determined because it is
overlapped by another element"
x1 "background color could not be determined because it partially
overlaps other elements" → .z-maker-line__text
```

That last one is axe independently noticing BLOCKING-1.

Because axe punted, I computed contrast myself (`qa/harness/contrast.mjs` — WCAG relative-luminance formula, effective background resolved by walking ancestors and compositing alpha, `aria-hidden` subtrees excluded, large-text threshold applied per element):

```
@1280: 206 text elements — FAILS: 0 — over a gradient: 0 — lowest
4.89:1 (needs 4.5)
```



```
@375 : 188 text elements — FAILS: 0 — over a gradient: 0 — lowest  
5.53:1 (needs 4.5)
```

Lowest measured pairs: `#8492A8` on `#1D2532` = 4.89:1 (`.z-trow__platform`, 15px/400) and `#8492A8` on `#141A24` = 5.53:1. All clear AA. Brand manual §4's co-render rule is satisfied — every state carries **colour + glyph + text label**, with the glyph `aria-hidden` so AT reads the label once:

```
○ NOT STARTED  rgb(165,162,155) = #A5A29B  ● IN PROGRESS  
rgb(255,176,0)  = #FFB000  
● ZERADO       rgb(25,224,255)  = #19E0FF  ○ ABANDONED  
rgb(199,125,255)= #C77DFF
```

Ratified order Not Started · In Progress · Zerado · Abandoned is intact, and chip 3 is composed differently exactly as §05 requires: width 336px vs 249px, glyph 30px vs 24px, and the page's **only** cyan glow.

3 · Horizontal overflow — ⚠ assertion passes, real overflow present

See MAJOR-1 and BLOCKING-1 above. Per-element check found 3 offenders at 375, 3 at 768, 1 at 1280, 0 at 1920; after excluding the `aria-hidden` watermark, all are the §02 paragraph and its `<strong>`.

Intentionally-scrollable containers: none are keyboard-trapped.

`TerminalFrame` never needed its `overflow-x: auto` backstop — at 375 the terminal frames reflow to two-line rows as §5.5 mechanism 2 intends, and no `.z-frame__screen` reported `scrollWidth > clientWidth`. The only `overflow: hidden` containers are `.z-frame__watermark-clip` (decorative, `aria-hidden="true"`) and the root. So there is no scrollable region requiring `tabindex="0"`, and none is missing.

4 · Full keyboard traversal — ✅ pass

Chromium, 1280 and 375.

- **Skip link is first in DOM and first on Tab**, `href="#main"`, target `#main` exists.
- **It becomes visible on focus.** ⚠ *Correction to my own first measurement:* I initially recorded the focused skip link at `top: -900px` (off-screen) and nearly filed it as blocking. That was a measurement artifact — I sampled computed style in the same tick as the 140ms `top` transition. Re-measured over time:

```
t=0ms    top=-892.406px  onScreen=false  ← what a naive probe
sees
t=50ms    top=-355.625px  onScreen=false
t=150ms    top=16px      onScreen=true
t=400ms    top=16px      onScreen=true
t=1000ms   top=16px      onScreen=true
@375px    top=16px      rect x=16 y=16 w=176  onScreen=true
```

Visually confirmed in `screenshots/skiplink-focused-375.png` — cyan pill, top-left, focus ring drawn.

- **It works.** Enter → `location.hash=#main`, scrollY 0→97; the next Tab lands on the hero CTA *inside* `#main`, correctly skipping the nav's links. (`#main` has no `tabindex="-1"`, so `document.activeElement` stays `body` and the browser's sequential-focus starting point does the work — functional in all three engines tested, though an explicit `tabindex="-1"` would be more robust.)
- **Focus order matches DOM order exactly** — 16 tab stops at 1280, 15 at 375, compared element-by-element against `querySelectorAll` DOM order. No divergence.
- **Every stop has a visible indicator:** all 16 report `:focus-visible` true with `outline: 2px solid rgb(25,224,255)`, `outline-offset: 2px` — constant geometry, as §6.2 specifies. Ring is present under `prefers-reduced-motion: reduce` too (only its transition collapses).
- **No focus trap** — Tab exits the document after the last stop.
- **No unreachable interactive element, and no zero-size focusable.** At 375 the nav GitHub link measures 0×0 — verified `display: none`, i.e. correctly removed from the tab order (15 stops, `zero-size focusable: 0`). This is the ratified §5.2 no-hamburger behaviour ("the GitHub link drops to the footer"), **not a defect**.
- `<details>` open and close by keyboard: 6/6 pass.

```
details[0..5] closed→false Enter→true Enter→false OK
```

5 • `prefers-reduced-motion: reduce` — pass, including the pip park

The critical assertion holds. Measured `.z-scanner-track--live .z-scanner-track__pip`:

```
no-preference : anim=z-scanner-sweep dur=2.4s iter=infinite
offset-from-track-centre=72px opacity=1
reduce : anim=none dur=0s offset-
from-track-centre=0px opacity=1
visibility=visible display=block pip=18% of track
bg=rgb(255,46,46) glow intact
```

The pip parks dead-centre at full opacity and is never hidden — blueprint §7.6's explicit requirement. Animated-element count across the page drops from 1 → 0; all transitions collapse to `0.001s`.

#	MOTION	REDUCED-MOTION STATE OBSERVED	VERDICT
1	ScannerRule sweep (1 live, hero)	<code>animation: none</code> , parked at centre offset 0px , opacity 1, visible	✓ per spec
2	Parked scanner tracks	<code>animation: none</code> in both modes — static by design	✓
3	CTAButton primary hover (3 found)	<code>.z-cta</code> 0.14s → 0.001s	✓
4	CTAButton secondary hover (1 found)	same <code>.z-cta</code> rule, 0.14s → 0.001s	✓
5	Focus ring appearance	ring still drawn (2px solid <code>rgb(25,224,255)</code> , offset 2px) in reduce; only transition collapses	✓ per spec
6	NavLink / FooterLink underline (4 links)	<code>.z-nav-link</code> , <code>.z-footer-link</code> ×2, <code>.site-footer__powered</code> 0.14s → 0.001s	✓
7	FAQItem expand (220ms spec)	measured 0s in both modes — never implemented	⚠ MINOR-1
8	FAQItem collapse (140ms spec)	measured 0s in both modes — never implemented	⚠ MINOR-1
9	GridHorizon	static in both modes	✓

`scroll-behavior: auto` in both modes. No scroll-triggered animation, parallax, marquee, or autoplay found.

6 • Cross-browser — ✓ pass, no meaningful divergence

Chromium 151.0.7922.34 · Firefox 153.0 · WebKit 26.5, each at all four viewports (`qa/qa-crossbrowser.json`).

- Zero page errors, zero 4xx/5xx responses, in every engine at every viewport.
- Overflow assertion result identical in all three (and BLOCKING-1 reproduces identically in all three).

- Document heights agree to within 0.4%: 1280 → chromium 11,442 / firefox 11,443 / webkit 11,442.
- `<details>` styling — consistent: `summary` height 56px in all three; `list-style: none` + `::-webkit-details-marker { display: none }` suppresses the native marker everywhere; the custom `+` / `-` marker renders in all three.
- `clamp()` — supported everywhere; H1 resolves to 30.0007px (Chromium) / 30px (FF) / 30.000675px (WebKit) at 375, and 52px in all three at 1280. H1 box identical at 335×97 (375) and 650×168 (1280).
- `aspect-ratio` — `CSS.supports` true in all three; cover tiles lay out identically (12 tiles, uniform grid).
- **Gradients** — 12 gradient-bearing elements in all three, no fallback divergence. **No text sits on a gradient** (independently verified: `over a gradient: 0`), so no contrast risk from them.
- `backdrop-filter` — supported in all three but **used by zero elements**; not a risk on this page.
- **Variable fonts** — all three self-hosted `woff2` files load in every engine (Orbitron 400–900, Space Grotesk 300–700, JetBrains Mono 100–800, `status: loaded`), served same-origin, HTTP 200. Fallback stacks (`Michroma`, `Arial Black`, `system-ui`) present.

Only divergence found, and it is benign — one extra wrapped text line in Firefox/WebKit at narrow widths:

```
@375  moods                chromium=1655  firefox=1683  webkit=1655
Δ=28px
@375  one-collection        chromium=1732  firefox=1760  webkit=1760
Δ=28px
@768  states                chromium=710   firefox=735   webkit=710
Δ=25px
@1280 and @1920: no section differs by more than 4px.
```

7 · Content honesty sweep — PASS, no violations

- **Forbidden tokens: zero occurrences.** Scanned both visible `innerText` and full serialised HTML for the Q4-banned community-source name, `lorem ipsum`, `TODO`, `[TBD]`, `{{`, `${`, `R$ 00`, plus `FIXME`, `XXX`, `placeholder`, `Lorem`, `undefined`, `NaN`. Result: `forbidden-token hits: NONE`. Lone-`X` placeholder regex over visible text: `NONE`. The Q4 constraint (the banned community-source name appears nowhere — copy, icon, screenshot or FAQ) **holds**.

- **All four roadmap phases read "Planned".** Rendered: `PHASE 1 CLI/TUI MVP ...`
`○ PLANNED`, `PHASE 2 Enrichment ... ○ PLANNED`, `PHASE 3 Recommendations & Budget ... ○ PLANNED`, `PHASE 4 Social & Mo... ○ PLANNED`. 4 `.z-status-marker` nodes, each `○ + Planned`, colour `#A5A29B` — none marked done.
- **\$11 community has zero interactive elements.** Queried `a[href], button, input, select, textarea, summary, [tabindex]:not([tabindex="-1"]), [role="button"], [onclick]` inside `#community` → **0 results**.
- **No donate/sponsor anywhere.** Searched text and HTML for `donate, sponsor, patreon, ko-fi, kofi, buymeacoffee, github sponsors, paypal, pix` → `money words found: []`, `donate/sponsor LINKS: []`. The two ratified money *statements* are both present and are disclosure only: the \$11 line *"servers cost money — so this part will need a premium account or a donation once it exists. Nothing is decided about price, tiers, or a date, and nothing on this page is asking..."* and the footer's affiliate-commission disclosure. Exactly the Q3/Q6 resolution in `decisions.md`.
- **All 6 mockup figures carry a visible `<figcaption>`**, each `role="img"` + `aria-labelledby` → its own caption id, each verbatim from `content/landing-copy.md`:

\$	CAPTION VISIBLE	OPENING WORDS
hero	✓	"An illustration of Zerado's terminal library view. This is a mockup of the planned interface, not a screenshot..."
moods	✓	"Mockup of Zerado's mood picker in the terminal..."
one-collection	✓	"Mockup of a cover-art grid... Cover tiles are illustrative artwork, not real game covers. "
terminal-first	✓	"Mockup of a game's detail page in the terminal..."
and-on-your-phone	✓	"An illustration of two phone frames... Not built yet — Phase 4."
community	✓	"Example mockup of the community layer... watermarked 'Phase 4, not available yet.'"

- **The cover-grid caption renders**, appended to the \$06 figcaption, verbatim: *"Cover tiles are illustrative artwork, not real game covers."*
- **"Powered by FlowForge" is present and correct:**
`href="https://www.flowforgesoft.com"`, `rel="noopener noreferrer"`, wrapping `img src="/flowforge-logo.svg" alt="FlowForge"` which **resolves 200** (`image/svg+xml`, natural 150×150). Not a 404. (See MINOR-6 re: 18px vs 28px render size.)

- Bonus checks against `decisions.md` Q5: *zerado* (25 mentions) and *sinopse* (2) are kept, each glossed in-line ("Portuguese for..."), and both first appearances are `<em>`-italicised per the copy casing rule.

8 • Link integrity — PASS, zero 4xx/5xx

10 `<a href>` on the page. Full inventory:

```
[0] #main "Skip
to content"
[1] https://github.com/JustCode-CruzAlex/Zerado rel=noopener
noreferrer nav
[2] mailto:alex@flowforgesoft.com?subject=Zerado-WaitList
nav
[3] mailto:alex@flowforgesoft.com?subject=Zerado-WaitList
$hero
[4] https://github.com/JustCode-CruzAlex/Zerado rel=noopener
noreferrer $hero
[5] mailto:alex@flowforgesoft.com?subject=Zerado-WaitList
$closing-cta
[6] mailto:alex@flowforgesoft.com?subject=Zerado-WaitList
$footer
[7] https://github.com/JustCode-CruzAlex/Zerado rel=noopener
noreferrer $footer
[8] mailto:alex@flowforgesoft.com
$footer contact
[9] https://www.flowforgesoft.com rel=noopener
noreferrer target=_blank $footer
```

- **In-page anchors:** 1 (`#main`) → target id exists. No `href="#"`, no dangling fragment.
- **Outbound HTTP** (curl, no redirect / following redirects):

```
https://github.com/JustCode-CruzAlex/Zerado → no-redirect:200
followed:200
https://www.flowforgesoft.com → no-redirect:200
followed:200
```

No CTA is dead. Outbound destination set is exactly the expected `{github, flowforgesoft.com, mailto}`.

- `mailto:` **subject verified:** all 4 waitlist CTAs carry `?subject=Zerado-WaitList` exactly; the footer contact link is deliberately bare per copy §16.
- **Same-origin assets, all 200:** `/`, `/fonts.css`, `/logo.svg`, `/flowforge-logo.svg`, `/favicon.svg`, `/_astro/index.af0tTXyW.css`, and 3 `woff2` files. `requestfailed` **count: 0**. Zero external requests — the §1.2 zero-external-request commitment holds.

9 · Semantic structure — pass

```
lang = "en"
title = "Zerado – terminal game library tracker, sorted by mood"
meta description present
landmarks: banner/header=1 main=1 contentinfo/footer=1 nav=1
headings: 13 h1 count: 1 skipped levels: NONE
```

Heading ladder is `h1` → `h2` ×12, one per section that the blueprint gives a heading; §02 `maker-line`, §15 `closing-cta` and §16 `footer` correctly carry none, matching §1.1's *"(no heading)"* entries.

Images: 2 total (the page ships zero raster images as §1.2 commits — every "image" is composed HTML/CSS or SVG). Both have `alt`: `/logo.svg` → `alt="Zerado"`, `/flowforge-logo.svg` → `alt="FlowForge"`. The six mockup figures use `role="img"` + `aria-labelledby` → visible `<figcaption>`, so their alt text is the on-screen caption and matches `content/landing-copy.md` verbatim (table in §7 above) — announced once, not twice.

JavaScript: one `<script>` tag, `type="application/ld+json"` (`schema.org SoftwareApplication`) — structured data, not executable script. The §1.2 zero-JavaScript commitment holds. Zero page errors in all three engines.

10 · Blueprint conformance — 16/16 sections, 38/40 components

All 16 sections present, in the ratified DOM order, with the exact ratified ids:

```
hero · maker-line · the-problem · moods · states · one-collection ·
price-intelligence ·
yours-and-offline · terminal-first · and-on-your-phone · community ·
roadmap ·
after-phase-4 · faq · closing-cta · footer
```


0 added · 0 dropped · 0 reordered (15 `<section>` in `<main>` + `<footer id="footer">`).

Component inventory (`qa/harness/components.mjs`) — 38 of 40 resolve to DOM nodes. Counts match spec where spec gives one: CTAButton primary ×3, secondary ×1, FAQItem ×6, TerminalRow ×6, MoodCard ×4, StateChip ×4, StoreRow ×4, PhaseCard ×4, StatusMarker ×4, TrustPoint ×3, PhoneFrame ×2, CoverGrid ×1, PriceCard ×1, Sparkline ×1, CommunityPanel ×1, GridHorizon ×1, `.z-tile` ×12. Two did not resolve: `StateLegend` (MINOR-5 — absent, and unplaced by the blueprint's own §05 spec) and `CoverTile` under my first guessed selector — it resolves as `.z-tile` (12 nodes), so it is present; only its variant count deviates (MINOR-4).

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§03 What I could NOT test, and why

1. **Real assistive technology.** No VoiceOver / NVDA / JAWS run. Everything in §9 is programmatic (roles, names, landmark and heading structure). Announcement *quality* — how the six `role="img"` figures and the glyph/label state pairs actually sound — is unverified. This is the largest residual gap.
2. **Real devices and real touch.** All four viewports were emulated in desktop engines at `deviceScaleFactor: 1`. No physical iOS/Android device, no real Safari-on-iOS (Playwright WebKit ≠ iOS Safari), no actual finger-target testing, no on-device font rendering.
3. **The `mailto:` hand-off itself.** I verified the four CTA `href`s carry `?subject=Zerado-WaitList` and that the address is well-formed, but I cannot launch a mail client to confirm a composer opens with the subject pre-filled. The `decisions.md` Q2 trade-off (mail-client hand-off converts worse; the address is scrapable) is a product decision, already recorded, not re-litigated here.
4. **Core Web Vitals, Lighthouse, bundle size.** Deliberately out of scope — the `performance` lane owns these and I was told not to duplicate. I report no LCP/CLS/INP number and none should be inferred from this document.
5. **Variable-font rasterisation quality.** I verified all three fonts load with the correct weight ranges in all three engines; I did not assess hinting, subpixel rendering, or glyph-level quality, which needs a human eye on real hardware.
6. **`forced-colors` / Windows High Contrast Mode, and `prefers-contrast`.** Not emulated. A dark-native page with no light mode is exactly the kind of surface where forced-colors can go wrong; worth a follow-up.
7. **400% zoom (WCAG 1.4.10's other half).** I tested the 320px-width reflow floor — where BLOCKING-1 is at its worst (50 characters lost) — but did not separately exercise browser zoom at 400%, which can expose different failures than a narrow viewport.
8. **axe's own contrast verdict.** axe returned `incomplete`, not `pass`, for 42 nodes. The clean contrast result in §2 is **my** implementation of the WCAG relative-luminance formula (`qa/harness/contrast.mjs`), not axe's. It is auditable and I have shown its numbers, but it is not an independent tool corroborating axe — the 31 glyph-only nodes remain formally unverified by axe, though they are all `aria-hidden="true"` decoration whose meaning is carried by an adjacent text label.
9. **One correction worth recording.** My first skip-link probe read `top: -900px` on focus and would have been filed as a second BLOCKING finding. It was wrong — the sample landed inside a 140ms transition. The re-measured time

series is in §4. No claim in this report rests on a single-tick computed-style read.

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§04 Bottom line

Ship-blocking work is **one CSS declaration in one component**. `.z-maker-line__text` must be allowed to shrink and wrap; that single fix clears BLOCKING-1 and MAJOR-2 together. Separately, and independent of the fix, the `overflow-x: hidden` on the document root should be reconsidered (MAJOR-1) — it is what let a 58%-content-loss defect pass the design's own prescribed gate assertion at every viewport.

Everything else on this page is in good shape: zero axe violations, zero contrast failures, zero dead links, zero placeholder or dishonest content, a correct and visible skip link, a fully keyboard-operable page with a consistent focus indicator on all 16 stops, an exactly-conforming reduced-motion implementation including the scanner pip park, and three browser engines that agree to within 28px on an 11,442px page.

Re-run everything with: `cd qa/harness && node run-core.mjs && node run-kbd.mjs && node run-rm.mjs && node run-content.mjs && node run-xb.mjs && node contrast.mjs`

ADDENDUM – fix round, verified by the orchestrator

Appended after the fix round. The verdict above (FAIL, 1 blocking) was correct when written; this records what changed and how it was proven. The original findings are left intact — a report that edits away its own findings can't be audited.

§05

BLOCKING-1
— §02

maker-
line

text
clipping ·

RESOLVED

Root cause, confirmed: `.z-maker-line__text` carried `flex-shrink: 0`, pinning it at its 762px intrinsic width forever. The flanking `.z-maker-line__rule` spans had `flex-basis: 0%`, so they could never receive any share of flex negative-space redistribution — which is why they rendered at 0px at every viewport. Two mechanisms, one trigger.

Fix: text → `flex: 0 1 auto; min-width: 0`; rules → `flex: 1 1 0%` with `min-width: 24px`, forcing the flex resolver to take the remainder from the text.

Independently re-measured by the orchestrator (Playwright, live preview, `#maker-line .z-maker-line__text`):

WIDTH	CHARS RENDERED	RIGHT EDGE / VIEWPORT	FLANKING RULES
320	78 / 78	252 / 320 ✓	24px · 24px
375	78 / 78	307 / 375 ✓	24px · 24px
768	78 / 78	692 / 768 ✓	24px · 24px
1280	78 / 78	948 / 1280 ✓	24px · 24px
1920	78 / 78	1268 / 1920 ✓	24px · 24px

Full string at 375: *"Zerado is the second product from FlowForgeSoft, the company behind FlowForge."* Character count matches `content/landing-copy.md` exactly at all five widths. The ratified FlowForgeSoft attribution now survives on a phone, and the rule-flanked layout renders as designed.

FlowForge	DRAWING TITLE					
	BLOCKING-1					
	— §02 maker-					
	line text					
	clipping ·					
RESOLVED				SHEET	DWG NO.	REV
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§06 The finding behind the the tautological gate finding — is CLOSED

This report's most valuable observation was that its own prescribed gate could never fail: `documentElement.scrollWidth === clientWidth` is pinned to `true` by `overflow-x: hidden` on the root, so it reported PASS at all four viewports while 58% of a section's text was being destroyed.

`qa/harness/run-core.mjs` has been changed by the orchestrator (backup at `run-core.mjs.bak`):

- `pass` is now `offenders.length === 0` — the per-element check the harness already computed every run but never gated on. The old document-level comparison is retained as `documentScrollEqualsClient` for reference, demoted from verdict to datum.
- `aria-hidden` subtrees are excluded from `offenders`, so the by-design clipped `PHASE 4` watermark no longer masquerades as content loss.

Mutation-tested, because a gate nobody has watched fail is just a different tautology:

```
375px HEAD (fixed): offenders=0 PASS
375px MUTANT (bug re-injected): offenders=2 KILLED → p.z-
maker-line__text, strong
768px HEAD (fixed): offenders=0 PASS
768px MUTANT (bug re-injected): offenders=2 KILLED → p.z-
maker-line__text, strong
```

The pair is what matters: **green at HEAD** and **red at the mutant**. The gate now detects the exact defect it previously slept through.

FlowForge	DRAWING TITLE				
	The finding behind the finding — the tautological gate is CLOSED				
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§07 Re-run against the fixed build —

run-
core.mjs

,
live

```
[375] overflow pass=true | axe violations=0 passes=28  
incomplete=1  
[768] overflow pass=true | axe violations=0 passes=28  
incomplete=1  
[1280] overflow pass=true | axe violations=0 passes=28  
incomplete=1  
[1920] overflow pass=true | axe violations=0 passes=28  
incomplete=1
```

The single reported scroller at every width is `.z-frame__watermark-clip` — `overflow-x: hidden`, not `auto`, inside an `aria-hidden` subtree. It is a clip, not a scrollable region, so it correctly needs no `tabindex`. Not a defect.

FlowForge

DRAWING TITLE

Re-run
against the
fixed build —
run-core.mjs,
live

SHEET

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§08

No-regression sweep (orchestrator, against the rebuilt

dist/)

CHECK	RESULT
Forbidden tokens (the Q4-banned source name, lorem, TODO, placeholder, {{)	0
Roadmap phases reading "Planned"	4
Powered by FlowForge footer block	1, correct href
Dead anchors (href="#")	0
mailto: carrying subject=Zerado-WaitList	4
Ratified sections present, in order	16 / 16
<h1> count	1
JS files in dist/	0
External hosts in built HTML	github.com · flowforgesoft.com · zerado.app · schema.org — no CDN, no analytics

DRAWING TITLE

No-regression
sweep
(orchestrator,
against the
rebuilt dist/)

FlowForge

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§09

Also fixed this round (from the performance lane, and one gap the orchestrator found)

- **Conditional mono preload** — `JetBrainsMono-latin.woff2` (42.7% of page weight) no longer races the LCP-critical fonts on mobile, where it has zero above-fold consumers. Verified by CDP capture: `isLinkPreload=false` at 375, `true` at 1280.
- **fonts.css inlined** — one render-blocking round trip removed; requests 8 → 7; desktop render-blocking savings now below Lighthouse's reporting threshold.
- **public/_headers added** — `immutable` only for content-hashed `/_astro/*`, revalidating policy for the *unhashed* fonts (avoiding a stale-font trap), `no-cache` for `index.html`.
- **OG image, found missing by the orchestrator** — the page declared `twitter:card=summary_large_image` with **no image**, so every social share would have rendered blank. A 1200×630 branded card is now built, wired via absolute `https://zerado.app/og-card.png`, with honest alt text: the card carries a visible *"MOCKUP OF THE PLANNED INTERFACE"* label and the alt text does not imply a real screenshot.

DRAWING TITLE

Also fixed this
round (from
the
performance
lane, and one
gap the
orchestrator
found)

FlowForge

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STATUS

● IN REVIEW

§ 10

Verdict after fix round: PASS

Zero blocking findings. The largest remaining gap is unchanged and worth restating plainly: **every accessibility check here is programmatic**. Real screen-reader announcement quality has not been verified by a human with a screen reader, and axe returning 0 violations is not the same claim.

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DRAWING TITLE

Verdict after
fix round:
PASS

SHEET

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STATUS

● IN REVIEW

*Verified empirically. The product
behaves as drawn.*

FlowForge
ENGINEERED ARTIFACT

DRAWING TITLE

Zerado landing page — QA report
(accessibility & cross-browser lane)

DISCIPLINE	SHEET	DWG NO.	REV	STATUS
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LEGEND ☐ Field — blueprint ☐ Ink — annotation ☐ Accent — registration ☐ Pass — verdict

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